

University of Hohenheim
AIDAHO – AI & Data Science Certificate
Introduction to Data Science with R and RStudio



Automatic Identification System (AIS)
Traffic Analysis and Vessel Behaviour

Take Home Assignment 2026

Team 40

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https://aidaho-edu.uni-hohenheim.de/gitlab/Jakoob2/aidaho_ids_thas_2026

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<http://193.197.229.211:7080>

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1 Overview of the AIS Data

1.1 AIS Background

The Automatic Identification System (AIS) is a maritime tracking system in which vessels broadcast information about their identity, position and movement. AIS messages are transmitted by onboard transponders and received by terrestrial base stations or satellites. The data are commonly divided into static vessel information, such as MMSI, IMO number, vessel name, flag, type and dimensions, and dynamic position information, such as timestamp, latitude, longitude, speed over ground (SOG), course over ground (COG), heading and rate of turn [1]. In this assignment, AIS data are accessed through the AIDAHO PostgREST API, which exposes selected database tables as REST resources.

1.2 API Resources and Data Model

The available API resources identified through the PostgREST endpoint are: `ais_static`, `ais_dynamic`, `ais_dynamic_daily_counts`, `ais_dynamic_daily_counts_mat`, `ais_germany`, `germany_h3`, `ingested_files`, `geography_columns` and `geometry_columns`. The main analytical tables are `ais_static` and `ais_dynamic`. The static table contains one row per vessel identified by `mmsi`, including vessel name, flag, ship type, call sign, draught, length and width. The dynamic table contains time-series AIS broadcasts with message timestamp, coordinates, speed, course, heading, navigation status and collection type. Both tables are linked by `mmsi`. Because `ais_dynamic` is very large, dynamic queries were restricted by time intervals, vessel identifiers, aggregation or limits.

1.3 Descriptive Statistics

The static vessel table contains 228,427 rows and therefore 228,427 distinct vessels. The dynamic AIS table contains approximately 826,793,024 rows, while the Germany-specific table `ais_germany` contains approximately 389,947,776 rows.

The `ais_static` table contains 13 recorded vessel types. A complete overview is shown in Table 1. The most frequent categories are *Other*, *Cargo*, *Fishing Vessel*, *Tanker*, and *Tug*. Commercial vessel categories such as *Cargo* ships and *Tankers* are expected to dominate because AIS is particularly relevant for maritime logistics, regulated shipping, and vessel monitoring. The large share of the broad *Other* category suggests that some AIS metadata are not fully standardised or cannot be assigned to a more specific class.

Table 1: Vessel types recorded in `ais_static`.

Vessel type	Count
Other	57,974
Cargo	51,781
Fishing Vessel	34,736
Tanker	16,496
Tug	13,257
Pleasure Craft	9,014
Special Craft	7,624
Passenger Ship	7,147
Sailing Vessel	5,164
High-Speed Craft	1,227
Search and Rescue	1,201
Wing In Ground	644
Reserved	320

Vessels operate under 229 distinct flags. The most frequent flag is China (CN) with 67,214 vessels, followed by the United States (US, 13,771), Panama (PA, 8,024), the Netherlands (NL, 7,635) and Norway (NO, 7,303). The high frequency of China and large shipping registries such as Panama is plausible given the global structure of merchant shipping and flag registration.

The `ais_static` table contains a mix of identifier, numeric, categorical and timestamp variables. For example, `mmsi` and `imo` are stored as integers, `draught`, `length` and `width` as numeric variables, while `name`, `flag`, `ship_type` and `destination` are text-based categorical variables. Temporal information such as `eta` and `static_updated_at` is stored as timestamps.

Table 2: Selected summary statistics for numeric variables in `ais_static`.

Column	Non-missing	Min.	Max.	Mean
<code>mmsi</code>	228,427	–	–	–
<code>imo</code>	121,121	–	–	–
<code>draught (m)</code>	121,121	0.0	25.5	4.72
<code>length (m)</code>	216,537	0	1,022	63.6
<code>width (m)</code>	216,537	0	126	11.5

Table 2 summarises selected numeric variables in `ais_static`. For identifier variables such as `mmsi` and `imo`, only non-missing counts are reported because averages are not meaningful. Zero values and the maximum vessel length of 1,022 metres suggest occasional metadata errors or missing AIS information [2].

For categorical variables, visible patterns indicate incomplete standardization of AIS metadata. The most frequent values in several text-based fields show that not all entries represent meaningful real-world information. In particular, vessel names, call signs and destinations contain repeated placeholder values, zero-like entries and special characters such as `@`. This suggests that some AIS metadata are manually entered, incompletely transmitted or inconsistently standardised [2].

Table 3: Examples of frequent categorical values indicating AIS metadata quality issues.

Column	Frequent value	Count	Interpretation
name	@@@@@@@@@@@@@@@@@@@@	666	Placeholder-like vessel name rather than a meaningful ship name.
call_sign	@@@@@@@	25,653	Very frequent missing or placeholder call sign.
call_sign	0000000	7,492	Zero-like call sign, likely not a valid operational identifier.
destination	@@@@@@@@@@@@@@@@@@@@	17,135	Placeholder destination value.
destination	SHANGHAI@@@@@@@@@@@@	593	Real destination text with trailing filler characters.

1.4 Filtered Query Results

The following results were obtained using PostgREST filters and HEAD counts where appropriate. For Benelux countries, vessels were filtered by Belgian, Dutch and Luxembourg flags using both country names and abbreviations. This returned 9,476 vessels.

German cargo ships were identified with filters for German flags and cargo ship types. The query returned 291 German cargo vessels, of which 76 were longer than 150 metres. A case-insensitive name filter for **EXPRESS** identified 572 vessels containing this term in their name.

For the first five minutes of 2022, the dynamic table was filtered from 2022-01-01 00:00:00 UTC to 2022-01-01 00:05:00 UTC. This snapshot contains 1,506 AIS records and 1,100 distinct vessels. During the interval from 2021-05-04 14:00:00 UTC to 2021-05-04 14:30:00 UTC, 76 vessels were moving faster than 12 knots. After matching these MMSIs with cargo vessels from **ais_static**, 31 of them were identified as cargo ships.

The detailed query logic was based on filters such as `flag=in.(Germany,DE)`, `ship_type=ilike.*cargo*`, `length=gt.150` and `name=ilike.*EXPRESS*`. Returned records can be restricted with the `limit` parameter, for example:

```
1 /ais_static?select=mmsi,name,flag,ship_type&limit=5
```

2 Sampling Strategies

2.1 Sampling Strategy and Representativeness

A sampling strategy is a systematic procedure for selecting observations from a larger population. In the context of AIS data, sampling is necessary because the `ais_dynamic` table contains hundreds of millions of observations and is too large for efficient interactive analysis. Sampling is therefore not simply the act of using less data. Instead, the goal is to create a manageable subset that still reflects the main patterns of the original population. A good sample should be reproducible, transparent and preserve important characteristics such as temporal, spatial and vessel-type variation [3].

A sample is representative if it reflects the underlying population and avoids systematic distortions that may bias conclusions [3]. AIS data can become non-representative in several ways. First, temporal bias may occur if only specific periods are sampled, potentially overrepresenting busy or quiet traffic phases. Second, spatial bias may arise when some regions have better receiver coverage than others. Third, vessel-type bias may occur if certain vessel classes or transmission behaviours are overrepresented in the sample.

Availability bias describes distortions caused by unequal data availability rather than the true underlying population. In AIS data, this can result from differences between terrestrial and satellite AIS receivers. Terrestrial receivers mainly cover coastal and inland areas, whereas satellite AIS captures offshore traffic but may perform worse in dense shipping regions. Consequently, coastal traffic may be overrepresented in terrestrial data, while remote offshore movements or smaller vessels can be underrepresented [1].

2.2 Interval-Based Cluster Sample

The first sample was based on all possible 5-minute intervals on 24 January 2024. Using the fixed seed 1103276, 100 intervals were randomly selected. For each interval, up to 100 AIS observations were retrieved, stopping once 10,000 observations had been collected. The resulting dataset was saved as `sample_intervals.csv`. Setting a fixed seed ensures that the randomly selected intervals can be regenerated later for validation or debugging.

However, the seed only controls random operations performed locally in R. It does not determine which rows a remote API returns when a query uses `limit` without a defined row order. During validation, this caused different machines to retrieve different records for otherwise identical requests. To make the API-based sampling more reproducible, dynamic AIS requests that return individual observations were therefore ordered explicitly by `msg_timestamp` and `mmsi` before applying the row limit. This makes the retrieved subset deterministic for a given time window, while the sampled time intervals remain controlled by the fixed seed.

This approach represents cluster sampling because complete time intervals act as clusters [3]. A possible bias is temporal cluster bias, where unusually busy or quiet periods may be overrepresented. Another issue is vessel activity bias, since vessels transmitting more frequently are more likely to appear in the sample. A stratified sample by `ship_type` and hour of day can reduce these biases by ensuring a more balanced representation across vessel types and time periods.

2.3 Stratified Sample

The second sample used a stratified sampling strategy based on `ship_type` and hour of day [3]. To avoid API size and runtime limits, the data were queried in smaller hourly intervals. In total, 336 strata were identified and their share of the total record count was calculated. The target sample size of 10,000 observations was allocated proportionally across strata, while very small non-empty strata received at least one observation. Since true random retrieval through the API was not possible, hourly subsets were retrieved in a deterministic order and then sampled locally

using the fixed seed 1103276. The resulting dataset was saved as `sample_stratified.csv`. This approximation avoids unfiltered requests to the large dynamic table and ensures that the same hourly extracts are used when the script is rerun.

2.4 Comparison of Samples

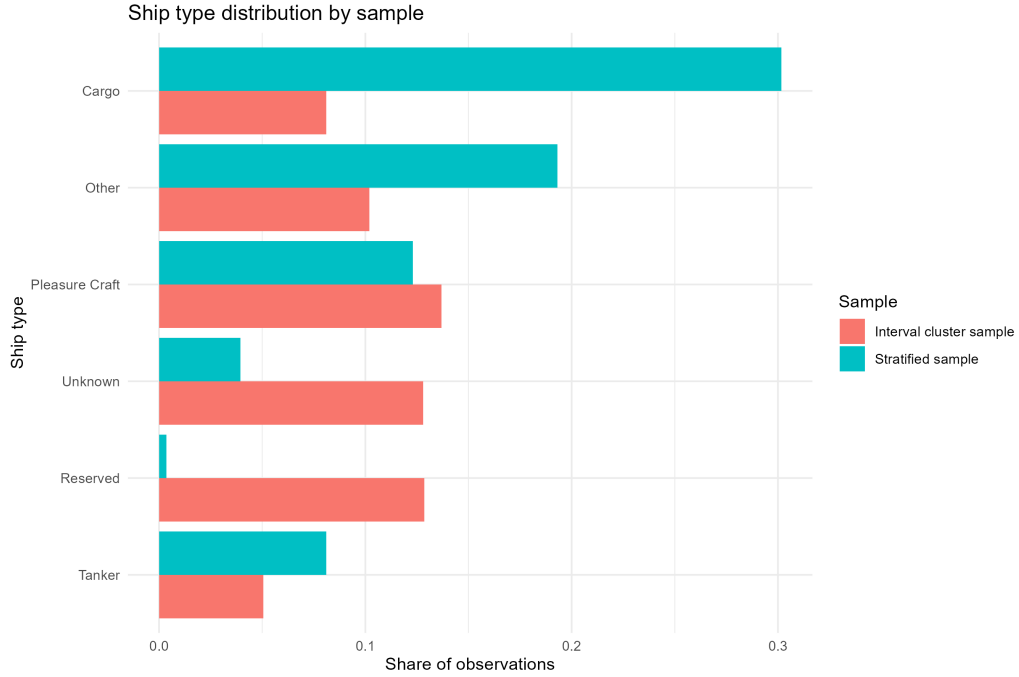


Figure 1: Comparison of ship-type distributions between the interval cluster sample and the stratified sample. Additional comparison plots for speed and collection type are provided in `03_report/graphs/`.

Figure 1 shows the main structural difference between both sampling approaches. The stratified sample contains a larger share of cargo and tanker vessels, while categories such as *Unknown* or *Reserved* occur less frequently. Additional comparison plots for speed and collection type are stored in `03_report/graphs/`. These plots show that the speed distributions are broadly similar, although the stratified sample contains slightly more medium-speed observations. Differences are also visible in collection type, where the stratified sample includes more `dynamic_classB` observations. These differences arise because the stratified sample explicitly balances observations across vessel types and time periods, whereas the interval sample depends more strongly on randomly selected time clusters.

3 Vessel Behaviour

3.1 Sample Dashboard Map

The interval-based sample from Task 3 was visualised as an interactive leaflet map. Each AIS observation is shown as a point, with vessel speed encoded by colour. Low speeds are represented by lighter colours, while darker colours indicate faster vessel movement. The legend therefore applies to individual AIS points and allows the user to interpret spatial traffic patterns together with vessel speed.

To keep the interactive HTML maps responsive, nearby AIS observations are grouped using leaflet marker clustering. The coloured cluster circles, including green, yellow or orange cluster symbols, do not represent vessel speed. Instead, the number inside each cluster indicates how many AIS observations are grouped at that location and zoom level. After zooming in, clusters split into individual AIS points, and only these individual points should be interpreted using the yellow-to-red speed scale.



Figure 2: Interactive leaflet map of the interval-based AIS sample. Individual point colour indicates vessel speed, with darker colours representing higher speed over ground. Cluster circles show the number of grouped AIS observations and do not encode speed.

3.2 Satellite-Only Map

A second version of the map was created by filtering the sample to observations with `collection_type = "satellite"`. Compared with the full sample, the spatial pattern changes because satellite AIS and terrestrial AIS cover different parts of maritime traffic. Terrestrial receivers mainly capture vessels close to coastlines, ports and inland waterways, while satellite AIS can also record vessels farther offshore. As a result, the satellite-only map emphasizes more open-water observations, whereas dense coastal or inland clusters are less dominant than in the full sample.



Figure 3: Satellite-only leaflet map of the interval-based AIS sample. The map only includes observations with `collection_type = "satellite"`. As in the full map, cluster circles show grouped observation counts rather than vessel speed.

3.3 Individual Vessel Tracks and Dynamic Paths

The script `ais_dynamic_individual_paths.R` was developed to query, process, analyse, and visualise individual vessel trajectories from the AIS database. To ensure efficient rendering of large numbers of AIS points in interactive leaflet maps, marker clustering was applied using `markerClusterOptions()`. Repeated query, processing, and plotting logic was implemented through helper functions to avoid redundant code and improve maintainability. Additional visual outputs for the inspected individual vessels are stored in `03_report/graphs/`. This includes interactive leaflet maps for MMSI 2579999, MMSI 563040400 and MMSI 211430830. The report focuses on the main lock-detection map, while the remaining HTML maps are provided in the repository for reproducibility and visual inspection.

3.3.1 MMSI 2579999: Suspicious AIS Behaviour

The AIS observations for MMSI 2579999 indicate clear data quality issues. The recorded positions span implausibly large geographic distances within a short time period, including observations across multiple continents. The resulting track therefore cannot be interpreted as the movement of a single normal vessel.

External AIS-related information suggests that this identifier should be treated with caution, as it may be associated with a navigation aid, base station, default configuration, or another special-purpose AIS transmission source rather than a conventional moving ship. This interpretation is also consistent with the observed data pattern: the positions show geographically inconsistent jumps that would be physically impossible for a vessel. Therefore, the observations are interpreted as a technical artefact or special-purpose AIS source, not as a realistic vessel trajectory [4].

3.3.2 MMSI 563040400: Vessel Movement near Constanța

The vessel with MMSI 563040400 was analysed for the time window from 2021-01-01 00:00:00 UTC to 2022-01-01 00:00:00 UTC. The queried observations form a coherent and geographically plausible track near the port of Constanța in the Black Sea.

The trajectory indicates typical vessel movement behaviour. Lower speeds and denser AIS observations occur near the harbour area, suggesting manoeuvring, waiting, or docking activity. In contrast, movement outside the port area appears smoother and faster, which is consistent with regular navigation. The vessel reached a maximum observed speed of approximately 11.7 knots.

3.3.3 MMSI 211430830: Vienna–Linz Route and Ship Locks

The movement of MMSI 211430830 was analysed for the period from 2021-01-17 00:00:00 UTC to 2021-01-20 00:00:00 UTC. During this period, the vessel travelled along the Danube between Vienna and Linz and passed several ship locks.

In AIS data, ship locks typically appear as local clusters of observations at nearly identical coordinates with very low speed. Since vessels must wait, enter the lock chamber, be raised or lowered, and leave afterwards, AIS points accumulate in the same area for several minutes. Compared with normal cruising segments, lock passages therefore exhibit reduced speed, increased observation density, and temporary pauses in the vessel trajectory. A noticeable gap in the AIS trajectory is visible between the Vienna region and the western section of the route. This likely reflects temporary missing AIS coverage or incomplete message reception rather than an actual interruption of the vessel journey. AIS observations may be missing due to terrestrial receiver coverage limitations, transmission interruptions, signal obstruction in narrow river valleys, or incomplete historical data ingestion. Since the vessel reappears later on the same river corridor with a plausible continuation of the trajectory, the missing segment is interpreted as a data gap rather than unusual vessel behaviour.

3.3.4 Rule-Based Ship Lock Detection

To automatically detect potential ship-lock passages, a transparent and reproducible rule-based heuristic was implemented. The detection algorithm consists of the following ordered decision steps:

1. Sort all AIS observations by `msg_timestamp`.
2. Mark observations as potential lock points if vessel speed is below 1 knot.
3. Group consecutive low-speed observations if they are both spatially and temporally close, defined as a maximum time gap of 30 minutes and a maximum spatial distance of 600 metres between consecutive points.
4. Summarise each group by start time, end time, duration, mean latitude, mean longitude, and number of observations.
5. Remove implausible groups shorter than 8 minutes or containing fewer than 3 AIS observations.
6. Interpret all remaining groups as potential ship-lock passages.

These thresholds were selected to provide a transparent and reproducible detection strategy. A speed threshold of 1 knot captures stationary or nearly stationary vessel behaviour typical for lock passages. The temporal and spatial constraints prevent unrelated waiting periods from being merged into the same event, while minimum duration and observation thresholds reduce false positives caused by isolated low-speed messages.

Using this heuristic, five potential lock events were identified during the observed route.

Table 4: Detected potential ship-lock passages for MMSI 211430830. Times are reported in UTC.

Lock ID	Start time	End time	Duration (min)	Latitude	Longitude	Obs.	Median speed (kn)
1	Jan 18 05:21:46	Jan 18 05:46:15	24.48	47.8826	17.5411	6	0.10
2	Jan 18 19:20:02	Jan 18 19:41:42	21.67	48.1762	16.4789	8	0.10
3	Jan 18 20:55:00	Jan 18 21:18:01	23.02	48.2437	16.3857	12	0.05
4	Jan 19 20:45:02	Jan 19 21:03:12	18.17	48.2502	14.4317	7	0.10
5	Jan 19 21:55:00	Jan 19 23:57:35	122.58	48.2781	14.3494	42	0.10

The rule-based approach is transparent and reproducible but also has limitations. Slow movement caused by anchoring, waiting areas, port congestion, or sparse AIS transmission may incorrectly be classified as lock passages. Similarly, short or poorly recorded lock transits may remain undetected. Potential improvements could include incorporating known ship-lock locations, river network constraints, heading changes, acceleration patterns, or spatial clustering methods such as DBSCAN [5].

An optional unsupervised alternative would be to cluster low-speed AIS observations in space and time without labelled training data. Algorithms such as DBSCAN or HDBSCAN could automatically identify dense stationary clusters along the vessel route and classify plausible clusters as potential lock passages.

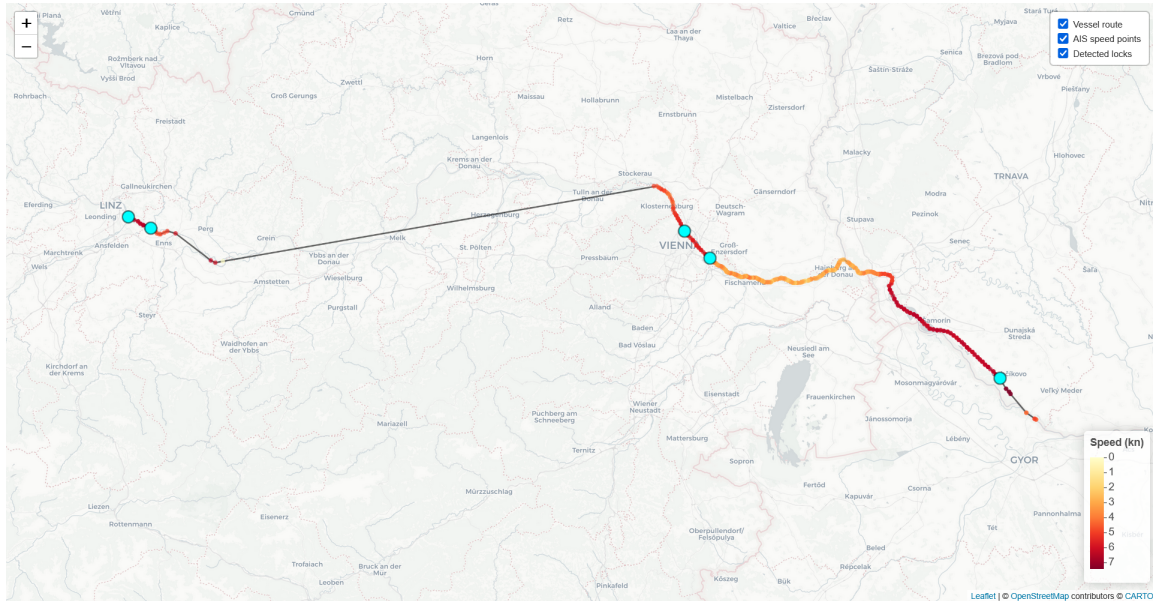


Figure 4: Trajectory of MMSI 211430830 between Vienna and Linz. Regular AIS observations are shown along the route, while turquoise points indicate detected potential ship-lock passages.

4 AIS Traffic Density on German Rivers

4.1 German River Geometries

German river geometries were retrieved with the `rnaturalearth` package from the Natural Earth `rivers_lake_centerlines` dataset. Only relevant German inland-waterway rivers were retained using a name-based selection of major rivers. Synonyms such as *Rhine/Rhein*, *Danube/Donau* and *Mosel/Moselle* were standardised, and duplicated geometries were dissolved by river name. The geometries were clipped to the German boundary so that only river segments within Germany were used for the later AIS matching.

For visual inspection, a map of the German river geometries was created. All retrieved river geometries are shown, but the legend is restricted to major rivers to keep the figure readable. This avoids an overcrowded legend while still showing the spatial structure of the inland waterway network.

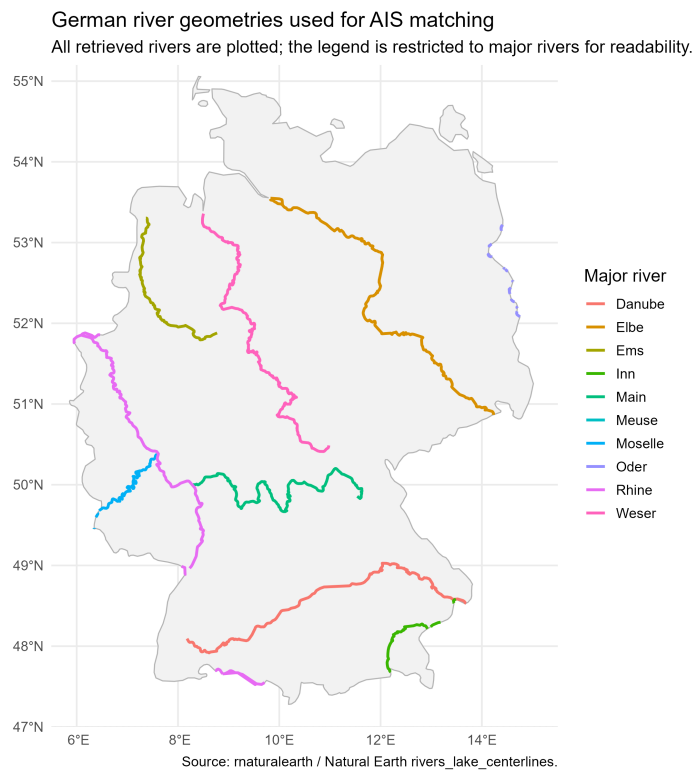


Figure 5: German river geometries used for AIS matching. All retrieved rivers are plotted, while the legend is restricted to major rivers for readability.

The `ais_germany` table was queried for the analysis date 2022-04-23. To avoid downloading unnecessary data, the query was restricted to records from this single day only. In total, 497,990 AIS records were retrieved and saved as `ais_germany_2022_04_23.csv`.

4.2 H3-Based River Matching

To determine whether AIS observations are located on German rivers, the H3 spatial indexing system was used. H3 divides the earth into hierarchical hexagonal grid cells, each represented by a unique index. Hexagonal grids provide more spatially uniform neighbourhoods than square grids, since each cell has six equally spaced neighbours and fewer directional distortions. This makes spatial relationships more stable and reduces edge effects compared with naive geometric matching approaches [6].

H3 is more efficient than comparing every AIS point with every river geometry directly, because the spatial matching problem can be reduced to matching H3 index keys. A hash table stores values using unique keys for fast lookup operations. In this context, H3 indices act as hash keys: both AIS points and buffered river cells are converted into H3 indices, allowing efficient matching through key comparison instead of repeated geometric distance calculations.

For each AIS record from 2022-04-23, an H3 index was calculated from latitude and longitude. The same H3 resolution was used for the river geometries. As required, H3 resolution 7 was selected. Since the river geometries are line features, a buffer of 1,500 metres was applied before generating the river H3 cells. This buffer accounts for generalized Natural Earth river centre lines, river width, GPS noise and small deviations between AIS positions and the mapped river geometry. If an AIS point matched more than one river, it was assigned to the nearest original river geometry.

In total, 497,990 AIS records were analysed for 2022-04-23. Of these, 258,108 records (51.83%) were assigned to selected German rivers, while 239,882 records (48.17%) were classified as elsewhere.

Table 5: AIS records assigned to selected German rivers on 2022-04-23.

River	AIS records
Rhine	117,007
Elbe	95,019
Weser	32,932
Main	5,424
Danube	3,526
Moselle	2,942
Ems	1,258

The Rhine and Elbe account for the largest number of river-associated AIS records, followed by the Weser. This reflects the strong role of these rivers in German inland shipping and port-related traffic.

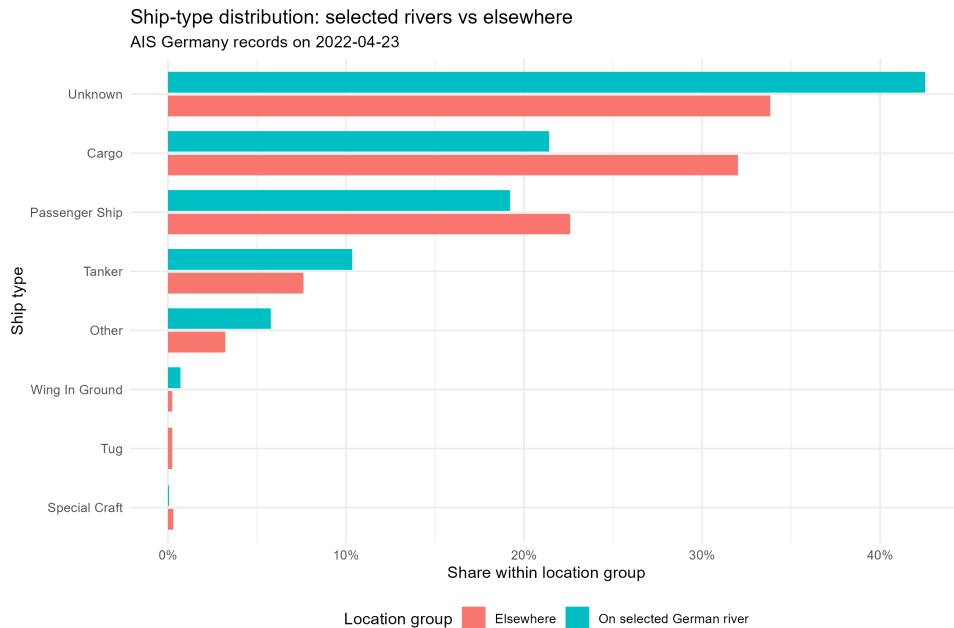


Figure 6: Grouped bar chart comparing ship-type distributions between AIS records on selected German rivers and records elsewhere.

Figure 6 shows that *Unknown*, *Cargo* and *Passenger Ship* dominate both location groups. River-associated records contain a higher share of *Unknown* and *Tanker* observations, while records elsewhere show a higher share of *Cargo* and *Passenger Ship* observations.

4.3 Traffic Density along the Rhine

For the Rhine density analysis, AIS records assigned to the Rhine were projected onto the Natural Earth Rhine geometry and aggregated into 10 km distance bins. The start point and direction follow the order of the Rhine geometry as stored in the Natural Earth dataset. Consequently, the x-axis represents relative distance along this stored geometry rather than an official hydrological flow direction or Rhine kilometre system.

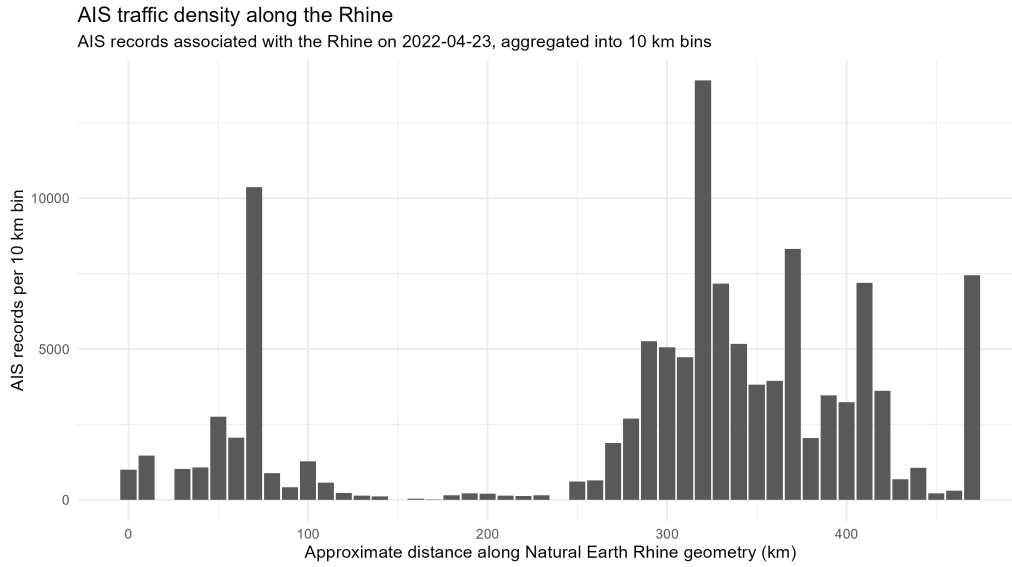


Figure 7: AIS traffic density along the Rhine on 2022-04-23, aggregated into 10 km distance bins.

Peaks in Figure 7 indicate sections with particularly dense AIS activity, while lower bars indicate less frequently observed river segments. The final distance bin may be elevated because observations near the end of the stored line accumulate in the last segment. Additional visual outputs from the river-matching and traffic-density analysis are stored in `03_report/graphs/`. The report includes the main river geometry map, the grouped ship-type comparison and the Rhine density plot, while additional intermediate plots are provided in the repository for reproducibility and visual inspection.

5 Deployment of the Dashboard and Shiny Application

5.1 Deployment Architecture

The leaflet dashboard was exported as a self-contained HTML file using `htmlwidgets::saveWidget()` and saved as `sample_points.html`. The file was placed in the NGINX document root via a Docker volume mount (`assets/site-content`), allowing it to be served as a static webpage.

The deployment uses Docker Compose with two services: an NGINX container for static content and reverse proxying, and a Shiny container for the interactive application. The Shiny app files are mounted from `assets/shiny` into the container, while preprocessed input data are loaded locally to avoid repeated expensive API requests during user interaction.

The Shiny application implements the river-density analysis from Task 5 interactively. Users can select a river and one or more ship types, then refresh the plot. The application displays AIS traffic density on 2022-04-23 aggregated into 10 km river-distance bins. NGINX routes requests to the Shiny service under `/ais_app/`. After local testing, the repository was updated on the assigned server and the containers were started using `docker compose up -d`.

5.2 Access to the Deployed Services

The deployed services can be accessed through the following URLs:

Static dashboard: http://193.197.229.211:7080/sample_points.html

Shiny application: http://193.197.229.211:7080/ais_app

Server entry page: <http://193.197.229.211:7080/>

AI Usage Statement

Generative AI tools (primarily ChatGPT) were used to support parts of this assignment, including code writing and structuring, debugging, query formulation, language refinement and LaTeX formatting. All generated content was critically reviewed, adapted where necessary and validated by the authors. Responsibility for the correctness and interpretation of all results remains with the authors.

References

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Reproducibility and Repository Information

For reproducibility and grading purposes, all scripts, generated data, deployment files and documentation are provided in the accompanying GitLab repository:

https://aidaho-edu.uni-hohenheim.de/gitlab/Jakoob2/aidaho_ids_thas_2026

The repository follows the folder structure required by the THAS guidelines and contains all code necessary to reproduce the reported results. A detailed explanation of the repository structure, execution order, required software, Docker deployment and Shiny setup is provided in the repository `README.md`.

The project uses `renv` for reproducible package management. Required package versions are documented in `renv.lock` and can be restored using:

```
1  renv::restore()
```

All figures included in the report are additionally stored in:

```
1  03_report/graphs/
```

The deployed web services referenced in the report remained accessible at the time of submission.